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Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

FULL STACK WEB DEVELOPMENT-CHRONIC KIDNEY DISEASE

AN INTERNSHIP REPORT

Submitted by,

ROHITHA S – 20221CSE0707

Under the guidance of,

Mr. Jerrin Joe Francis

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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BONAFIDE CERTIFICATE

Certified that this report **"Full Stack Web Development - Chronic Kidney Disease"** is a bonafide work of **Rohitha S (20221CSE0707),** who has successfully carried out the internship work and submitted the report for partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING during 2025-2026.

Mr. Jerrin Joe Francis	**Dr. Balakrishnan .R**	**Dr. Md Ziaur Rahaman**
Internship Guide	Program Internship Coordinator	School Internship Coordinator
PSCS	PSCS	PSCS
Presidency University	Presidency University	Presidency University

Dr. Asif Mohammed	**Dr. Duraipandian N**
Head of the Department,	Dean
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Presidency University	Presidency University

Name and Signature of the Examiners

Sl.No	Name of the Examiner	Signature/Date
1	Mr. H Senthil Kumar	27-04-26
2	Ms. Attar Mahay Sheetal	27/04

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DECLARATION

I am a student of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING , at Presidency University, Bengaluru, named ROHITHA S, hereby declare that the internship work titled "**Full Stack Web Development – Chronic Kidney Disease** " has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

Name of the Student: ROHITHA S USN: 20221CSE0707 Signature :

PLACE:BENGALURU

DATE: 27 - April 2026

ABSTRACT

Chronic Kidney Disease (CKD) is a widespread long-term health condition that often goes unnoticed in its early stages, making it especially dangerous. Many patients do not experience clear symptoms until the disease has significantly progressed, which highlights the importance of early detection in preventing severe kidney damage and improving outcomes. To address this challenge, a web-based CKD prediction system was developed using machine learning techniques to provide quick and reliable risk assessments based on basic clinical data. The system is built on the CKD dataset, which includes important health indicators such as blood pressure, specific gravity, albumin levels, red blood cell count, hemoglobin, and other biochemical parameters. A comprehensive data preprocessing pipeline was implemented to ensure model effectiveness, including handling missing values, encoding categorical variables, and normalizing continuous features. For predictive modeling, two widely used machine learning algorithms—Logistic Regression and Random Forest—were trained and evaluated.

The results showed that the Random Forest model significantly outperformed Logistic Regression, achieving around 100% accuracy compared to approximately 88%. This demonstrates the strength of ensemble methods in capturing complex patterns within medical data. To make the solution practical and user-friendly, the trained Random Forest model was integrated into a full-stack web application. The backend was developed using Flask to manage prediction logic, while the frontend was built with React to provide an intuitive interface. Through this platform, users can input patient data and instantly receive CKD risk predictions. Overall, the project showcases how machine learning can enhance early diagnosis, support clinical decision-making, and contribute to better healthcare outcomes.